

Energy, Heat and Work

Energy units, work and the first law

Name _____ Class _____ Date _____

EQUATIONS AND CONSTANTS

$$E_k = \frac{1}{2}mv^2 \cdot E_p = mgh \cdot g = 9.81 \text{ m/s}^2 \cdot w = -P(\text{ext})\Delta V \cdot \Delta U = q + w$$

$$1 \text{ cal} = 4.184 \text{ J} \cdot 1 \text{ kcal} = 4.184 \text{ kJ} \cdot 1 \text{ kWh} = 3.6 \times 10^6 \text{ J} \cdot 1 \text{ L}\cdot\text{atm} = 101.325 \text{ J}$$

Sign convention: q and w are positive when energy enters the system. Show units in every step.

A. Ideas

- (a) Define energy, heat and work. (b) Classify each as an open, closed or isolated system: a pan of boiling water with no lid; a sealed balloon; hot soup in a sealed, perfectly insulated flask.

- A bath of warm water (40 °C) and a cup of boiling water (100 °C). Which has the higher temperature? Which contains more thermal energy? Explain.

B. Kinetic and potential energy

- Calculate the kinetic energy of: (a) a 2.00 kg ball moving at 3.00 m/s (b) a nitrogen molecule (mass 4.65×10^{-26} kg) moving at 515 m/s.

- How much potential energy does a 5.00 kg box gain when it is lifted 2.00 m?

C. Energy units

5. Convert: (a) 520 kcal to kJ (b) 1200 kJ to kcal (c) 2.5 kWh to J

D. Work and the first law

6. A gas is compressed from 10.0 L to 4.0 L by a constant external pressure of 2.00 atm. Calculate the work in joules and explain its sign.

7. Calculate ΔU for each system: (a) loses 300. J of heat while 150. J of work is done on it (b) absorbs 500. J of heat and does 200. J of work (c) a gas in a rigid, sealed container absorbs 75 J of heat.

E. Challenge

8. A gas in a cylinder with a piston absorbs 2.50 kJ of heat and expands from 1.00 L to 6.00 L against a constant external pressure of 1.00 atm. Calculate (a) the work done, in J (b) ΔU , in kJ. (c) Did the internal energy of the gas increase or decrease?